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Fiscal Faux Pas?: An Analysis of the Revenue Implications of Trade Liberalization

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Summary. — This paper examines the argument that trade liberalization depresses tax revenue/GDP ratios in developing countries. This occurs because the structural characteristics associated with developing countries limit their ability to make the transition from trade to domestic taxes. Using a panel of 80 developing and industrialized countries over 1970–98, the econometric analysis carried out employs a fixed-effects regression framework to examine the evidence. The results indicate that low-income and upper middle-income countries have experienced declining tax revenues as a result of falling income and trade tax revenues and that structural characteristics have been significant in explaining the decline.

Key words — crosscountry, developing countries, fiscal reform, trade liberalization

1. INTRODUCTION

The rush to globalize during the past decade and a half has been necessitated predominantly by the severe internal and external imbalances which developing countries have experienced. It was and is widely believed that insufficient openness is primarily responsible for the crises in developing countries and hence, opening the economy constitutes the backbone of a sustainable growth process.

One aspect of external integration is the removal of trade restrictions. For many countries, however, trade interventions serve the dual purpose of protection and revenue generation. Efforts to liberalize trade will result in revenue losses in most cases, unless the liberalizing countries successfully replace the foregone revenue from trade by revenue from domestic sources. This paper argues that in many developing countries the transition from trade taxes to domestic taxes is not always possible because certain structural characteristics limit their ability to do so. In addition to low income, these features include large subsistence sectors, and heavy dependence on the taxation of trade for revenues and on primary product exports. Moreover, these economies are predominantly rural and have high ratios of dependents to working-age population.

Substantial empirical work has already been carried out on the determinants of revenue, in particular trade revenue. Most studies, however, focus on the volume of trade and the level of economic development as determinants, rather than the effects of trade liberalization on trade tax revenue. Greenaway (1984), Hitiris (1990), Ram (1994), and Tanzi (1987) have all found a positive relationship between the level of imports and import duties and trade tax revenue, as well as a negative relationship between the level of economic development (as measured by per capita income) and trade tax revenue. These studies have attempted to provide evidence for Kuznets' hypothesis that reliance on trade taxes declines as a country becomes more economically developed.

Studies examining the effect of macroeconomic policies on trade revenue have also been carried out. For instance, Nashashibi and Bazzoni (1994) find that in 28 countries of sub-Saharan Africa, devaluation of the exchange rate, declining terms of trade, and import liberalization undermined the tax base. A recent study by Ebrill, Stotsky, and Gropp (1999) measures the impact of trade liberalization on trade revenue using panel data and concludes that tariff reforms have not resulted in declining

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trade revenue. On the contrary, Rao's (1999) study on the effects of changes in openness (trade taxes relative to trade) and changes in the tax base (trade relative to GDP) on overall trade tax revenue, concludes that low-income countries overall, and countries of sub-Saharan Africa in particular, have confronted a tradeoff between reduced protection and reduced revenues from liberalization.

The econometric analysis in this paper uses panel data for 1970–98 for 80 countries. Using a fixed-effects regression framework, it examines the effects of trade liberalization on overall tax revenues. It also explores the nature of the relationship between the rate of trade taxation and trade revenue, specifically whether the rate of trade taxation was prohibitively high before the implementation of reforms and the effects that the easing of trade restrictions has had on customs revenues. The paper also examines how the structure of taxation changes with the level of development and openness.

The results show that developing countries, particularly low and upper middle-income countries, have suffered declining tax revenues as a result of falling income tax and trade tax revenues. Structural characteristics have been significant in explaining the decline in low-income countries. Moreover, before the implementation of reforms, all countries in the sample were below the revenue-maximizing rate of trade taxation. Hence, liberalization has imposed substantial fiscal costs.

The theoretical links between trade liberalization and tax revenue are discussed in Section 2. Section 3 examines the characteristics associated with unsuccessful fiscal adjustment. Changes in the tax structure at different levels of development, as well as changes in the tax structure arising as a result of greater openness, are also explored in this section. The determinants of tax revenue are investigated in Section 4, while Section 5 closes with some concluding remarks.

2. THE THEORETICAL FRAMEWORK

Developing countries have traditionally used trade taxes and subsidies, import and export quotas, and other nontariff barriers to regulate international trade. These have been used to achieve multiple goals; namely, raising revenue for the public sector, correcting market distortions, providing protection for local industry, improving the terms of trade by favorably affecting world market prices, and improving the distribution of income at home. Trade taxes have also been used in several instances to achieve exogenously determined levels of production and consumption. [1]

Neoclassical theory, however, contends that trade interventions diminish the efficiency of resource allocation, reduce competitiveness, and exacerbate macroeconomic imbalances (Krueger, 1995). Furthermore, it is argued that the static and dynamic gains from trade contribute to growth (see Krugman, 1990). Even if countries realize gains from trade, what are the implications for public finance?

Studies on public finance and taxation theory include trade taxes along with domestic taxes in the analysis of optimal taxation (Stern, 1987). The optimization problem consists in implementing tax policies which maximize the social welfare function subject to the constraints of technology, trade, and domestic tax revenue. In this framework, taxes on trade are viewed as creating both production and consumption distortions and are only part of the optimal tax structure because of collection cost considerations. [2] The political economy of public finance and trade theory furthermore argues against trade barriers since the very possibility of erecting such barriers leads to competitive lobbying by interested groups which, in turn, dissipates the very rents that the barriers create (Bhagwati, 1982; Krueger, 1974). [3]

The general conclusion in the literature is that the distortions created by trade taxes render them inferior to other fiscal policy instruments. Hence, if revenue considerations threaten to jeopardize trade reform, the domestic tax structure should be reformed to diminish reliance on trade tax revenues. [4] A shift to domestic income and consumption taxes will be more efficient, less inequitable, and easier to administer.

(a) *Problematic transition*

Policy options developed for advanced economies cannot, however, be transported to low-income countries indiscriminately as certain salient features of low-income countries frequently make the transition problematic. Furthermore, a well-designed tax system often must meet contradictory goals. In view of both multiple goals and the paucity of instruments, second-best considerations assume prominence to render certain policy options infeasible or suboptimal. Hence, the public finance literature does not give generally useful policy conclusions for developing countries. It is clear that optimality of the tax structure is a complex and contextual matter.

(i) *Structural constraints*

It has been observed that as an economy grows, it generally becomes more urbanized, as formalized in the Lewis two-sector model of structural change (Lewis, 1954). Urbanization increases both the need for tax revenues and the capacity to tax. On the demand side, greater urbanization leads to a greater need for public services. On the supply side, urbanization leads to larger taxable bases as economic activity tends to be concentrated in urban areas (Tanzi, 1987). [5]

Unlike urban activities, rural economic activities are difficult to tax because they are conducted in dispersed small-scale enterprises given to subsistence consumption rather than commercial production. Even where the rural economic surplus is substantial, it often escapes taxation. Moreover, because of the informal nature of economic activity, income bases are notoriously difficult to assess. Hence, governments must typically resort to taxing incomes in the traditional rural sectors by levying taxes on agricultural exports. Thus, countries heavily dependent on the exports of primary products tend to tax trade heavily. Developing countries also tend to have high age-dependency ratios, rendering income tax bases smaller than in more developed economies. All these features limit the range of available fiscal instruments.

(ii) *Institutional constraints*

Institutional factors too play a significant role in the collection of tax revenues. First, developing countries tend to have archaic tax administrations, making the assessment of tax liabilities problematic. This, compounded by the inclination of the underpaid tax administrators to be corrupt and the tendency of taxpayers to evade taxes, results in suboptimal revenue collection. Trade taxes, on the other hand, are relatively easy to assess and enforce, as monitoring the entry and exit of goods into and from the country is generally straightforward. Second, there exist political impediments that dislodge attempts to expand domestic tax bases. For instance, lobbying by politically powerful interest groups often results in their being exempt from taxation (Ndikumana, 2001).

(iii) *Contradictory objectives of modern tax systems*

Frequently, attempts to make tax systems more efficient beget further inefficiency. For instance, many developing countries have increased the taxation of domestic goods and services (by the imposition of sales taxes, value added taxes, and excise duties). But, as Tanzi (1989) points out, "much of the domestic production escapes taxation" and as a result a large proportion of sales taxes are imposed on imports. Excises too are predominantly levied on imported goods like petroleum, automobiles, alcohol, etc., diminishing the efficiency rationale for domestic taxation. Tax concessions made to interest groups also violate efficiency cannons. Moreover, in many countries revenues have increased rather dramatically after a large bout of privatization (Easterly, 1999; Edwards, 1996). These countries have relied on capital receipts arising from the sale of public enterprises, rather than implementing deep reforms, also contrary to the spirit of greater efficiency.

The efficiency component of fiscal systems is often orthogonal to equity considerations. A desirable feature of tax systems is the stress on redistribution of income and wealth by introducing an element of progressivity. The shift to domestic indirect taxes will promote a more equitable distribution of the burden of taxation if high rates are imposed on luxuries and low rates on necessities. In many poor countries, however, the mounting need for resources has resulted in taxation of necessities like food and fuel, placing a disproportionate burden on the poor. At the same time such taxation limits the consumption of a vast segment of the population which subsists on very meager incomes.

Domestic taxation is also administratively costlier. According to the World Bank (1988), administrative costs of customs and excise duties range from 1% to 3% of revenue collected as compared to approximately 5% for the VAT and 10% for personal income taxes. [6] It should be noted that these are measured average costs of taxation; marginal costs of domestic alternatives may well be considerably higher.

(b) *Revenue repercussions of trade liberalization*

The macroeconomic policies deployed during trade liberalization have varying effects on tax revenues and the net effect is often uncertain (Blejer & Cheasty, 1990; Tanzi, 1989). The process usually consists of the relaxation of quantitative restrictions, reduction and unification of tariffs, export promotion, and devaluation of the exchange rate, accompanied by domestic tax reform. These involve an important fiscal element as they influence the prices at which agents make consumption and production decisions; the ultimate effect on revenues depends on their demand and supply side responses to price changes. This section investigates the impact of policy reforms on tax revenues by analyzing the direct and indirect effects of trade reform in terms of the changing prices facing consumers and producers of exportables, importables and nontradables.

(i) *Tarrification of quantitative restrictions*

The replacement of quantitative restrictions by tariffs augments tax revenue receipts. As compared to quotas, however, tariffs increase the variability of domestic prices which could have adverse incentive effects on the volume of trade, diminishing revenue gains.

(ii) *Tariff liberalization*

Tariff liberalization involves reduction of the effective tariff rate, movement towards a uniform tariff rate on all imports, and phasing out of tariffs. According to the Laffer curve, if the initial tariff rate is prohibitively high, then tariff reductions necessarily increase revenue by reducing the incentive to evade taxes. Alternatively, if the initial tariff rate is below the revenue-maximizing rate, then the direct effect of a tariff reduction is loss of revenue.

The indirect effects are more difficult to assess and depend on the elasticity of substitution between imports and domestic import substitutes. When tariff rates are reduced, consumers tend to switch from domestic nontradables to imports or import substitutes, as the relative price of importables to nontradables declines. This reduces revenues obtained from taxation of nontradables and increases revenues from taxation of domestic import substitutes. The net change in tax revenues cannot, in general, be predicted. Moreover, with a decline in tariffs, the domestic production of nontradables and exportables increases, as their relative prices are higher. Income taxes on the income derived from the production of nontradables and exportables should now increase.

(iii) *Export promotion*

Trade liberalization also involves the reduction and removal of export taxes. If export taxes do not significantly change the volume of exports, their reduction should reduce export revenues.

(iv) *Terms of trade shocks*

When several developing countries liberalize simultaneously, they could face lower export prices due to an excess supply of similar exports, worsening their terms of trade. This would adversely affect revenues directly through reduced export revenues, or indirectly through the lower income earned from exports and hence lower income tax receipts.

(v) *Exchange rate adjustment*

Trade reform often entails devaluation of the real exchange rate. In a small open economy a real devaluation has several direct and indirect effects. First, devaluation causes the domestic currency value of imports to rise and assuming no change in import volumes, customs receipts will increase. It is more likely however, that the demand for imports declines and its magnitude is dependent on the price elasticity of import demand. Second, devaluation will result in expenditure switching from tradables to nontradables (as importables and exportables consumed domestically become more expensive). Thus, trade tax revenues should decline and revenues from domestic indirect taxes should rise. Devaluation also results in an increase in the production of tradables (as foreign demand for exports should increase and producing import substitutes is more profitable). Income taxes from tradables should therefore rise.

(vi) *Domestic tax reform*

Consumption taxes on nontradables might induce consumers to switch to tradables, thereby reducing potential revenues from domestic indirect taxes. But, if domestic consumption taxes are also levied on tradables, then indirect tax revenues should rise. Paradoxically, income taxes could decline as a result of efforts to increase them because the incentive to evade taxes might induce a proliferation of activity in the informal sector.

Consider the following identity for government tax revenues:

$$\mathrm{TR} = \mathrm{XR} + \mathrm{MR} + [\mathrm{DRT} + \mathrm{DRNT}] + [\mathrm{PYR} + \mathrm{KYR}] \qquad (1)$$

where TR is total government tax revenue, XR export revenue, MR import revenue, and

DRT + DRNT revenue from domestic taxes on goods and services. These are disaggregated into DRT, revenue from the taxation of tradables and DRNT, revenue from taxation of nontradables. PYR + KYR are total income tax receipts; PYR is personal income tax revenue, while KYR is corporate income tax revenue. TR can be further disaggregated into its components as follows:

$$\begin{aligned}\mathrm{TR} = {} & t_{\mathrm{e}}(P_{\mathrm{X}}^{*}eX) + t_{\mathrm{m}}(P_{\mathrm{M}}^{*}eM) + t_{\mathrm{d}}(P^{*}e\mathrm{DT}) \\ & + t_{\mathrm{d}}(P)(\mathrm{DNT}) + t_{\mathrm{y}}(\mathrm{PY}) + t_{\mathrm{y}}(\mathrm{KY}) \qquad (2)\end{aligned}$$

where t_{e} is the effective tax rate on exports, P_{X}^{*} the world price for exports, e the nominal exchange rate, X the volume of exports, t_{m} the effective tax rate on imports, P_{M}^{*} the world price of imports, M the volume of imports, t_{d} the effective tax rate on domestic goods and services, P^{*} the effective world price for tradables, DT the volume of tradables consumed domestically, P the domestic price level, DNT the volume of nontradables, t_{y} the effective income tax rate, PY the level of personal income, and KY the level of corporate income.

Taking the total differential of Eq. (2) and collecting terms yields:

$$\begin{aligned}\mathrm{dTR} = {} & \mathrm{d}t_{\mathrm{e}}(P_{\mathrm{X}}^{*}eX) + \mathrm{d}t_{\mathrm{m}}(P_{\mathrm{M}}^{*}eM) \\ & + \mathrm{d}t_{\mathrm{d}}(P^{*}e\mathrm{DT} + P\,\mathrm{DNT}) + \mathrm{d}t_{\mathrm{y}}(\mathrm{PY} + \mathrm{KY}) \\ & + \mathrm{d}e(t_{\mathrm{e}}P_{\mathrm{X}}^{*}X + t_{\mathrm{m}}P_{\mathrm{M}}^{*}M + t_{\mathrm{d}}P^{*}\mathrm{DT}) \\ & + \mathrm{d}P_{\mathrm{X}}^{*}(t_{\mathrm{e}}eX) + \mathrm{d}P_{\mathrm{M}}^{*}(t_{\mathrm{m}}eM) \\ & + \mathrm{d}P^{*}(t_{\mathrm{d}}e\mathrm{DT}) + \mathrm{d}P(t_{\mathrm{d}}\mathrm{DNT}) \\ & + \mathrm{d}X(t_{\mathrm{e}}P_{\mathrm{X}}^{*}e) + \mathrm{d}M(t_{\mathrm{M}}P_{\mathrm{M}}^{*}e) \\ & + \mathrm{dDT}(t_{\mathrm{d}}P^{*}e) + \mathrm{dDNT}(t_{\mathrm{d}}P) \\ & + \mathrm{dPY}(t_{\mathrm{y}}) + \mathrm{dKY}(t_{\mathrm{y}}) \qquad (3)\end{aligned}$$

If trade liberalization has a neutral effect on revenues, $\mathrm{dTR} = 0$. On closer inspection of Eq. (3), it can be concluded that $\mathrm{d}t_{\mathrm{e}}(P_{\mathrm{X}}^{*}eX)$ and $\mathrm{d}t_{\mathrm{m}}(P_{\mathrm{M}}^{*}eM)$ will be negative as the effective export tax rate and the effective tariff rate have declined, while $\mathrm{d}e(t_{\mathrm{e}}P_{\mathrm{X}}^{*}X + t_{\mathrm{m}}P_{\mathrm{M}}^{*}M + t_{\mathrm{d}}P^{*}\mathrm{DT})$ will be positive as the exchange rate has depreciated and thus $\mathrm{d}e$ is positive. But, prior expectations about the remaining terms cannot be formed rendering the net effect of trade liberalization on tax revenues ambiguous.

3. CROSS COUNTRY EVIDENCE ON THE REVENUE IMPACT OF TRADE LIBERALIZATION

The 80 countries used for the empirical analysis are classified into four income groups using the guidelines suggested by the World Bank, *World Development Report 2000/2001* (Appendix A). The database is constructed using data from the World Bank, *World Development Indicators* (2000) and various issues of IMF, *Government Finance Statistics*. The aim is twofold. First, the principal characteristics of unsuccessful fiscal adjustment and second, the structure of taxation and its relation to the level of development and openness are studied. The degree of trade liberalization is proxied by the effective rate of trade taxation, measured by the ratio of international trade taxes to the volume of total trade. [7] Hence, a drop in this ratio is indicative of greater trade openness.

(a) *Characteristics of unsuccessful fiscal adjustment*

Table 1 crossclassifies real per capita income with some economic indicators. 1980–84 is considered to be the point of departure for the analysis as it reflects the zenith of the economic crisis for most countries. This is compared with 1995–98 to assess results after almost a decade of reforms.

The main conclusions emerging from Table 1 are:

(i) Poorer countries on average are more closed than richer countries, even though all country groups have become more open since the mid-1980s.

(ii) Tax revenue/GDP ratios vary positively with the level of development, while the share of trade taxes in GDP varies inversely with the level of development.

(iii) More developed countries have lower age-dependency ratios, higher urbanization rates, and lower dependency on primary product exports than less developed countries.

(iv) Low-income and upper middle-income country groups have faced declining tax revenue/GDP ratios in the wake of trade reform.

(v) Falling revenues are associated with a substantial increase in the debt/GDP ratio in low-income countries, while indebtedness has declined in upper middle-income countries.

Table 1. *Structural characteristics and fiscal adjustment*[a]

Income group	Time period	TT	TR	OPEN	TRADE	PX	URB	DEP	DEBT
Low	70–74	4.88	12.14	11.78	48.22	n.a.	18.29	91.33	25.74
	75–79	4.90	12.96	10.90	52.09	n.a.	20.43	92.55	34.86
	80–84	5.02	14.52	10.15	53.00	81.36	22.66	92.81	45.73
	85–89	4.98	13.88	11.14	51.12	68.14	24.81	92.06	65.22
	90–94	4.86	13.83	9.20	56.27	65.49	27.10	90.55	78.94
	95–98	4.90	14.10	9.44	62.82	55.18	29.34	88.29	118.89
Lower-middle	70–74	5.05	14.21	9.21	56.66	n.a.	36.70	92.69	28.18
	75–79	6.05	16.80	9.67	66.32	n.a.	39.28	87.13	32.56
	80–84	5.27	17.16	7.44	68.98	79.86	41.76	81.92	61.93
	85–89	4.59	17.29	6.67	69.02	68.72	44.53	77.33	80.86
	90–94	4.81	17.81	6.06	82.84	58.51	47.38	72.22	51.91
	95–98	3.80	18.16	5.08	89.80	57.79	49.96	66.44	47.79
Upper-middle	70–74	4.67	14.88	6.87	61.21	n.a.	53.87	80.35	23.65
	75–79	4.30	18.57	5.37	70.49	n.a.	56.77	74.92	27.43
	80–84	4.11	19.71	5.33	74.64	61.93	59.71	70.05	47.30
	85–89	3.73	19.82	4.97	74.75	54.59	62.99	65.93	37.16
	90–94	3.56	19.82	4.41	79.02	47.55	66.00	61.91	32.76
	95–98	2.29	19.22	2.76	85.94	43.04	68.19	58.54	30.00
High	70–74	1.66	24.51	2.79	67.11	n.a.	71.27	59.36	30.04
	75–79	1.32	26.47	2.12	77.06	n.a.	72.99	56.83	38.43
	80–84	1.07	28.45	1.49	84.58	39.38	74.14	53.68	53.13
	85–89	0.75	29.39	1.13	82.22	35.19	75.20	51.29	57.00
	90–94	0.46	29.29	0.72	80.78	32.88	76.40	50.62	54.65
	95–98	0.25	30.54	0.38	83.65	31.10	77.59	50.37	56.59

Source: Authors' computations.

[a] All ratios are unweighted averages for the relevant period; n.a.: data not available; TT: trade taxes (% of GDP); TR: tax revenue (% of GDP); OPEN: trade taxes (% of trade); TRADE: trade (% of GDP); PX: primary product exports (% of total merchandise exports); URB: degree of urbanization; DEP: percent of dependents to working-age population; DEBT: central government debt (% of GDP).

Table 1 confirms the inverse relationship between the level of development and dependence on trade taxes as a source of revenue. During 1995–98, the share of trade taxes in GDP ranged from almost 5% in low-income countries to 0.25% in high-income countries. Evidence for other well-established empirical facts is also provided. As countries become more developed their tax revenue/GDP ratio rises, indicating that the capacity of the state to tax increases. During 1995–98, low-income countries had a tax revenue/GDP ratio of 14.10%, while high-income countries had a tax revenue/GDP ratio of 30.54%.

Low-income countries also exhibit distinct structural characteristics. They are less urbanized, have higher age-dependency ratios, and are more dependent on exports of primary products than the other country groups. As hypothesized, trade liberalization leads to the expected fiscal outcomes. As low-income countries have become more open and increasingly integrated with the international economy, the tax revenue/GDP ratio has declined by approximately 3% (from 14.52 to 14.10) between 1980–84 and 1995–98. This has had the effect of reducing public spending on physical as well as social capital. But, overall expenditure as a share of national income has remained roughly constant despite the conditionalities imposed on many developing countries by the international financial institutions to reduce fiscal deficits. This has transpired because of the rather substantial increase in interest spending (Khattry, 2001). As a result, indebtedness has increased considerably (by 160%). Surprisingly, greater openness has also led to falling revenues in the group of upper middle-income countries. Clearly, factors other than structural characteristics have been responsible for this weakness in revenues. As Khattry (2001) shows, the onus of the fiscal constraint has fallen on expenditures (in particular physical capital and defense) in this country group and therefore, indebtedness has not increased.

By contrast, lower middle-income countries have demonstrated successful fiscal adjustment after trade liberalization. They have reduced reliance on trade taxes and at the same time increased their tax revenue/GDP ratios. [8] They have also reduced their level of indebtedness, as debt/GDP ratios have declined from approximately 62% in 1980–84 to 48% in 1995–98. As Boote and Thugge (1997) explain, this decline has occurred largely as a result of the concerted efforts of the international lending community to help developing countries manage their debt.

(b) *Structure of taxation*

(i) *Structure of taxation and development*

As Table 2 indicates, income taxes as a share of GDP rise with the level of development. Direct income taxes contribute far less revenue than indirect taxes in developing countries. This is probably indicative of the inability to raise income tax revenues and of the high degree of tax evasion. For low-income countries, during 1995–98 the income tax/GDP ratio was only 3.83%, while the share of income taxes in tax revenue was 27.16%. In lower middle-income countries the income tax/GDP ratio was higher, at 4.79%. But, the share of income taxes in tax revenue was 27.30%, approximately the same as in low-income countries. Income taxes contribute progressively more (both as a share of GDP and of tax revenues) in the upper middle-income and high-income country groups.

The extreme disparities between groups concerning dependency on trade taxes is borne out by Table 2. While trade taxes contributed almost 35% of tax revenue in low-income countries, this share was less than 1% in high-income countries. Domestic indirect taxes do not exhibit any such distinct relationship. Although low-income countries had the lowest share of these taxes in GDP (4.66%) and high-income countries the highest share (10.20%)

Table 2. *Structure of taxation and level of development*[a]

Income group	Time period	YTAX	YTAX-TR	DTAX	DTAX-TR	TT	TT-TR	IND	IND-TR
Low	70–74	3.47	28.58	3.18	26.19	4.88	40.20	8.06	66.39
	75–79	3.53	27.24	3.82	29.48	4.90	37.81	8.72	67.28
	80–84	4.16	28.65	4.50	30.99	5.02	34.57	9.52	65.56
	85–89	3.60	25.94	4.59	33.07	4.98	35.88	9.57	68.95
	90–94	3.53	25.52	5.00	36.15	4.86	35.14	9.86	71.29
	95–98	3.83	27.16	4.66	33.05	4.90	34.75	9.56	67.80
Lower-middle	70–74	3.57	24.37	4.06	29.09	5.05	34.99	9.11	64.08
	75–79	4.64	27.32	4.15	26.06	6.05	34.70	10.20	60.76
	80–84	5.20	29.82	4.55	27.80	5.27	30.03	9.82	57.83
	85–89	5.27	29.64	5.28	31.76	4.59	26.33	9.87	58.09
	90–94	5.27	29.79	5.77	34.07	4.81	25.16	10.58	59.23
	95–98	4.79	27.30	7.40	41.85	3.80	20.18	11.20	62.03
Upper-middle	70–74	5.57	32.06	4.27	25.81	4.67	26.39	8.94	52.30
	75–79	6.06	31.26	4.73	24.63	4.30	21.96	9.03	46.59
	80–84	6.16	29.46	5.91	32.39	4.11	19.97	10.02	52.36
	85–89	6.41	30.43	5.90	31.69	3.73	18.25	9.63	49.94
	90–94	6.18	30.19	5.84	31.26	3.56	16.92	9.40	48.18
	95–98	5.98	31.10	7.56	39.37	2.29	13.04	9.85	52.41
High	70–74	8.81	35.94	7.73	31.54	1.66	6.77	9.39	38.31
	75–79	9.52	35.97	7.94	30.00	1.32	4.99	9.26	34.98
	80–84	10.36	36.41	8.67	30.47	1.07	3.76	9.74	34.24
	85–89	10.74	36.54	9.69	32.97	0.75	2.55	10.44	35.52
	90–94	10.37	35.40	9.62	32.84	0.46	1.57	10.08	34.41
	95–98	10.80	35.36	10.20	33.40	0.25	0.82	10.45	34.22

Source: Authors' computations.

[a] All ratios are unweighted averages for the relevant period; YTAX: taxes on income, profits, and capital gains (% of GDP); YTAX-TR: taxes on income, profits, and capital gains (% of tax revenue); DTAX: domestic taxes on goods and services (% of GDP); DTAX-TR: domestic taxes on goods and services (% of tax revenue); TT-TR: trade taxes (% of tax revenue); IND: indirect taxes (% of GDP) (DTAX + TT); IND-TR: indirect taxes (% of tax revenue) (DTAX-TR + TT-TR).

during 1995–98, the share of domestic indirect taxes in tax revenue was approximately one-third in both country groups. Lower middle-income countries are observed to raise more revenues from domestic indirect taxes than upper middle-income countries as a share of tax revenue, while domestic indirect taxes as a share of GDP are approximately similar in both middle-income country groups.

While indirect taxes are a far more important source of revenue in developing countries than direct income taxes, these taxes are equally important in high-income countries. Moreover, it is observed across all income groups that as the share of trade taxes in GDP declines, the share of domestic taxes on goods and services rises.

(ii) *Structure of taxation and openness*

Table 3 summarizes changes in the structure of taxation between 1980–84 and 1995–98. Several interesting observations can be made.

The decline in the tax revenue/GDP ratio in low-income countries has occurred not only as a direct consequence of declining trade taxes, but also as a result of declining income taxes. This has probably transpired not only because structural characteristics have prevented the transition to income taxes, but also because changing market incentives have led to greater tax evasion and unemployment, causing income tax receipts to decline. It is also observed that while trade taxes declined, domestic indirect taxes increased in low-income countries. Overall the net increase in total indirect taxes was 0.42%. Perhaps this occurred because trade taxes on tradables were replaced by domestic taxes on tradables.

Although the tax revenue/GDP ratio has increased in lower middle-income countries, the share of direct income taxes in national income has declined by approximately 8%, while the share of trade taxes in national income has decreased by approximately 28%. Domestic taxes on goods and services have increased substantially. Overall, the increase of 14% in the indirect tax revenue/GDP ratio has led to the observed rise of 6% in the tax revenue/GDP ratio between 1980–84 and 1995–98.

The tax revenue/GDP ratio has declined in upper middle-income countries largely because of declines in income tax and trade tax revenues. Income taxes as a share of national income have declined by about 3% since the mid-1980s, while domestic taxes on goods and services have risen, but have not been able to offset the decline in trade taxes. Consequently, indirect taxes as a share of GDP have experienced an overall decline of approximately 1.7% between 1980–84 and 1995–98.

While high-income countries were fairly open in the mid-1980s, changes in their tax structure demonstrate a greater movement towards efficiency. The increase in tax revenue has arisen mainly because of increases in income and domestic indirect taxes. Trade taxes have declined considerably, and the decline has been more than offset by increases in domestic taxes on goods and services.

4. DETERMINANTS OF TAX REVENUE

The determinants of tax revenue are empirically analyzed by regressing the share of tax revenue in GDP on the natural log of real per capita GDP (pcGDP), the natural log of population size (pop), the age-dependency ratio (dep), the degree of urbanization (urb), and the index of openness (tt). Population size and per capita income are entered in logarithms, while the remaining variables are entered linearly because the tax revenue function is assumed to be nonlinear in the scale of the economy (as measured by both population size and per capita income). Thus, as per capita GDP and population size rise, the tax revenue/GDP ratio increases, but at a decreasing rate:

$$\mathrm{TR}_{\mathrm{it}} = a_0 + a_1 \ln \mathrm{pop}_{\mathrm{it}} + a_2 \ln \mathrm{pcGDP}_{\mathrm{it}} + a_3 \mathrm{dep}_{\mathrm{it}} + a_4 \mathrm{urb}_{\mathrm{it}} + a_5 \mathrm{tt}_{\mathrm{it}} + e_{\mathrm{it}} \quad (4)$$

Table 3. *Structure of taxation and openness (percent change between 1980–84 and 1995–98)*

Income group	TR	YTAX	DTAX	TT	IND
Low	−2.89	−7.93	3.56	−2.39	0.42
Lower-middle	5.83	−7.88	62.64	−27.89	14.05
Upper-middle	−2.49	−2.92	27.92	−44.28	−1.70
High	7.35	4.25	17.65	−76.64	7.29

Source: Authors' computations.

The regression analysis includes fixed effects to account for unobservable country-specific and time-specific factors. That is, differences between countries are incorporated in the model by using dummy variables for individual countries and years.

Drawing on the discussion in Section 2, the level of income and the degree of urbanization are predicted to be positively correlated with the share of tax revenue in GDP, while the age-dependency ratio is expected to be negatively correlated with the tax revenue/GDP ratio. To control for scale effects, population size is included as a regressor. Population size and tax revenues should be positively correlated as a larger population size will raise the tax revenue/GDP ratio if there are economies of scale in tax collection (e.g., due to fixed administration costs). Finally, the index of openness and the tax revenue/GDP ratio should also be positively correlated because a more open economy should experience declining tax revenues. [9]

The results are reported in Table 4. [10] The only significant variables are population size, the urbanization rate, and the degree of openness. As hypothesized, the tax revenue/GDP ratio increases with population size and urbanization. It is observed that openness has led to declining tax revenue/GDP ratios: a drop of one percentage point in the effective rate of trade taxation results in a drop of 0.33 percentage points in the tax revenue/GDP ratio. Although the age-dependency ratio and the level of per capita income were statistically insignificant, their signs indicate that the tax revenue/GDP ratio increases as per capita income increases and declines as the age-dependency ratio increases.

We now examine the effect of openness on customs revenue by estimating the following equation.

$$\begin{aligned}\mathrm{TT_{it}} = a_0 + a_1\mathrm{tt_{it}} + a_2\mathrm{tt_{it}^2} + a_3\mathrm{ER_{it}} \\ + a_4 \ln \mathrm{pop_{it}} + a_5 \ln \mathrm{pcGDP_{it}} \\ + a_6\mathrm{dep_{it}} + a_7\mathrm{urb_{it}} + a_8\mathrm{trade_{it}} \\ + a_9\mathrm{dtax_{it}} + e_{it}\end{aligned} \quad (5)$$

Table 4. *Determinants of tax revenues*[a]

	All countries		LY		LMY		UMY	
	TR	TT	TR	TT	TR	TT	TR	TT
ln pop	2.93**	2.33*	4.80	4.66**	1.99	−0.88	−5.34	1.27
	(1.51)	(0.53)	(4.61)	(2.29)	(2.51)	(1.62)	(8.25)	(2.73)
ln pcGDP	0.20	0.10	4.81*	1.92**	−1.15	−1.31***	2.49	0.25
	(0.61)	(0.28)	(1.72)	(0.87)	(1.56)	(0.73)	(1.86)	(0.69)
dep	−0.04	0.01	−0.27*	0.02	−0.06	0.05	−0.04	0.01
	(0.03)	(0.01)	(0.07)	(0.05)	(0.04)	(0.03)	(0.12)	(0.04)
urb	0.13*	−0.05*	0.38***	−0.17***	0.23*	0.06	0.06	−0.05
	(0.05)	(0.02)	(0.21)	(0.10)	(0.08)	(0.06)	(0.12)	(0.04)
tt (open)	0.33*	0.77*	0.38*	0.75*	0.55*	1.06*	0.35*	0.35*
	(0.02)	(0.03)	(0.04)	(0.07)	(0.04)	(0.06)	(0.07)	(0.07)
tt^2		−0.01*		−0.01*		−0.02*		−0.01**
		(0.00)		(0.00)		(0.00)		(0.00)
trade		0.03*		0.05*		0.05*		0.03*
		(0.00)		(0.01)		(0.01)		(0.01)
dtax		−0.07*		−0.04		−0.17*		−0.17*
		(0.02)		(0.08)		(0.06)		(0.04)
ER		0.00		0.00		0.00		−0.00
		(0.00)		(0.00)		(0.00)		(0.00)
N	1755	1652	402	339	440	440	290	276
Adjusted R^2	0.83	0.81	0.78	0.91	0.89	0.92	0.88	0.90

*99% level.

**95% level.

***90% level.

[a] LY: low-income country group; LMY: lower-middle-income country group; UMY: upper-middle-income country group; TR: tax revenue (% of GDP); TT: trade tax revenue (% of GDP); ln pop: natural logarithm of population size; ln pcGDP: natural logarithm of real per capita GDP (1995 US$); dep: percent of dependents to working-age population; urb: degree of urbanization; tt (open): index of openness; trade: trade (% of GDP); dtax: domestic taxes on goods and services (% of GDP); ER: exchange rate (local currency per US$, period average); *N*: number of observations. Robust standard errors are reported in parenthesis.

where TT is trade tax revenue expressed as a percentage of GDP. The independent variables are the same as in Eq. (4) with the addition of the trade/GDP ratio (trade), the share of domestic indirect tax revenue in GDP (dtax), and the exchange rate (ER). The regression controls for population size, structural characteristics, and the level of trade. The relationship between the effective rate of trade taxation and trade tax revenue is assumed to be nonlinear as prohibitively high rates of trade taxation may lead to declining trade revenue. Hence, a quadratic form is used to estimate the effects of trade liberalization on customs revenue. The revenue-maximizing rate of trade taxation is obtained by solving for tt in the following equation: $a_1 + 2a_2(\mathrm{tt}) = 0$, i.e., $\mathrm{tt} = -a_1/2a_2$. The variable dtax has been introduced to examine the extent of the negative correlation observed between the trade tax/GDP and domestic indirect tax/GDP ratios in Section 3(b).

The results are reported in Table 4. The large positive coefficient of tt is indicative of a rather large tradeoff between reduced customs revenue and reduced protection. The negative magnitude of tt^2 suggests that a potential "Laffer effect" exists for trade tax revenues. But the revenue-maximizing rate of trade taxation is estimated to be 38.5% while, from Table 1, it is clear that the effective rate of trade taxation (OPEN) was well below this rate in all country groups during 1970–84. Thus, all country groups have been operating on the rising part of the Laffer curve.

Robust support is provided for the presence of a significant negative relationship between the trade tax revenue/GDP and domestic indirect tax revenue/GDP ratios. As dtax increases by a percentage point, TT declines by 0.07 percentage points. The results also indicate that a country whose trade/GDP ratio is one percentage point higher has a trade tax/GDP ratio that is 0.03% higher. Significant positive scale effects (as measured by population size) are seen to exist. Moreover, there is strong evidence that structural factors play a significant role in determining the trade tax revenue/GDP ratio across countries, as an increase of one percentage point in the urbanization rate is observed to lead to a decline of 0.05 percentage points in the dependent variable.

Four main results deserve emphasis. First, strong statistical evidence supports the hypothesis that reduced protection adversely affects overall tax revenue and trade revenue receipts. Second, the degree of urbanization exerts a significant positive influence on the tax revenue/GDP ratio and a significant negative influence on the share of trade tax revenue in national income. Third, the scale of the economy (as measured by population size) significantly and positively influences both the trade tax revenue and tax revenue/GDP ratios. Finally, the domestic indirect tax revenue/GDP ratio and the trade tax revenue/GDP ratio are significantly and negatively correlated, indicating that customs duties on tradables are partially replaced by domestic indirect taxes on tradables.

While the above results hold on average for the aggregate sample of 80 countries, the crosscountry variation could be large within the group of developing countries. Thus, we estimated Eqs. (4) and (5) separately for the low, lower-middle, and upper-middle-income groups using fixed effects to account for both unobservable country and time-specific factors. The results are reported in Table 4.

In keeping with expectations, structural factors exert the strongest influence on the tax revenue/GDP ratio in the low-income country group. Rising levels of per capita income, falling age-dependency ratios, and increasing urbanization, all lead to rising shares of tax revenue in national income. Structural characteristics continue to exert a significant influence on the tax revenue/GDP ratio in the lower-middle-income country group. But, now the degree of urbanization is the sole structural factor to significantly and positively influence the dependent variable. In the upper-middle-income country group, structural characteristics seem to play no significant role in influencing the tax revenue/GDP ratio. In all three developing country groups, greater openness has led to falling tax revenue/GDP ratios, while population size has not exerted any significant influence.

The trade tax revenue/GDP ratio has been significantly and positively influenced by both the population size and the level of per capita income in the low-income group. The presence of a significant positive correlation between the level of per capita income and the dependent variable is surprising, as we had expected a negative relationship. Perhaps this is accounted for by the fact that because income levels are so low in this group of countries, higher income facilitates increased trade and thus higher trade tax revenue. While the age-dependency ratio was not significant, it did have the expected positive coefficient. The urbanization rate has the expected significant negative coefficient, implying that an increase of one percentage point in the degree of urbanization will lead to a decline of 0.17 percentage points in the trade tax revenue/GDP ratio. A potential Laffer effect is seen to exist for customs revenue and the revenue-maximizing rate of trade taxation, estimated to be 37.5%, is higher than the actual rate. Higher trade/GDP ratios are seen to lead to rising shares of trade tax revenue in national income. Although a negative relationship is seen to exist between domestic indirect tax revenue and trade tax revenue, it is insignificant.

While structural characteristics such as the age-dependency ratio and urbanization rate do not significantly influence the trade tax revenue/GDP ratio in lower-middle-income countries, the level of per capita income exerts a significant and negative influence in keeping with expectations. Moreover, the results support the presence of a significant negative relationship between the domestic indirect tax revenue and trade tax revenue/GDP ratios. Support for a revenue-maximizing rate of trade taxation is also present. This rate is calculated to be 26.5%. Rising trade/GDP ratios are also observed to significantly and positively influence the trade tax revenue/GDP ratio.

In the upper-middle-income group of countries, structural characteristics and the level of development have no significant influence on the trade tax revenue/GDP ratio. Increased trade openness is observed to have led to declining trade tax revenue. The revenue-maximizing rate of trade taxation is estimated to be 17.5%. Like other developing country groups, increased trade has led to increases in the dependent variable. As for the lower-middle-income group, a significant negative correlation exists between domestic indirect taxes and trade taxes.

Thus, we find that the influence of structural factors on both trade tax revenue and tax revenue is the most pronounced in the low-income group. This influence progressively declines as the level of development (measured by the level of real per capita GDP) increases, to the extent that structural characteristics seem to play no significant role in explaining variations in tax revenue and trade tax revenue in the upper-middle-income group of countries.

5. CONCLUDING REMARKS

While increased openness might have relieved the external constraints to development (by increasing access to external capital, technology, manufactured imports, as well as export markets), it has worsened the internal constraints. Developing countries, especially low-income economies, have been constrained by the lack of tax instruments (stemming from low levels of per capita income, high age-dependency ratios, and low rates of urbanization) and shackled by their burgeoning debt, in the face of globalization. Reduced state capacity has consequently compromised the ability of the state to pursue development objectives.

As Musgrave (1969) has pointed out, the lack of available tax handles constrains revenue generation at low levels of income. But these constraints should lessen as an economy becomes more developed. The evidence in this paper supports this view and implies that policies which emphasize internal structural changes will aid countries more in the development process, rather than policies which rely exclusively on external integration to bring about structural change and development. As Chenery's (1979) work on the patterns of structural change indicates, these include a shift from agricultural to industrial production, a change in consumer demand from basic necessities to manufactured goods and services, concomitant with the growth of cities and urban industries (as people migrate from rural areas), a diversification of exports to include simple manufactured goods, and a decline in family size and in the population growth rate.

Not only do structural characteristics prevent countries from tapping domestic tax bases, but the shift to a market-based system of production, by changing economic incentives, actually causes declines in income tax/GDP ratios. Whether this has occurred as a result of the redistributive effects of greater openness or as a result of greater tax evasion, remains an open question. This is true of all three developing country groups, which have experienced declining income tax revenues in addition to falling trade tax revenues since the onset of trade liberalization.

The presence of inappropriate market incentives such as trade restrictions, has underpinned the policy prescriptions of the Bretton Woods institutions. These prescriptions, especially trade liberalization, were expected to alleviate the fiscal constraint over the medium term by improving efficiency. The evidence in this paper suggests that, on the contrary, the revenue constraint remains binding even after a decade of reforms. For low-income countries in particular, the revenue constraint has been compounded by an explosion of the debt/GDP ratio. Thus, the need for a fiscally realistic development strategy has become even stronger in the post-liberalization world.

NOTES

1. See Dixit and Norman (1980) and Farhadian-Lorie and Katz (1989) for discussions of the effects of trade restrictions.

2. See Bliss (1987) for elaboration of this point.

3. In neoclassical welfare economics lump-sum direct taxes provide the policy instruments for achieving the first-best, i.e., Pareto efficient, result. But lump-sum taxes are often impossible to design or implement and thus income and commodity taxation become necessary, giving rise to the theory of the second-best.

4. See, among others, Papageorgieu, Michaely, and Choksi (1991), and Rajaram (1994) who caution that greater emphasis on revenue issues is required during trade liberalization.

5. Increased urbanization might however, be accompanied by a proliferation of small-scale enterprises and a growth of the informal sector, which typically escape the tax net. Hence, urbanization might not result in the sought for revenues (Rao, 1999).

6. The same study also asserts that the economic costs associated with trade taxes are much higher than the economic costs associated with domestic taxes. But the economic costs associated with various taxes are calculated on the assumption that potential alternative tax instruments exist. If there are no such alternatives, then the calculations are invalidated and so too the conclusion.

7. Several criticisms could be directed against this index of openness. It could be argued that this index ignores the Laffer Effect by underestimating the effect of extremely high tax rates that result in little revenue and ignores the role of smuggling and other practices undertaken to evade taxes. In addition, it might depict an economy as becoming more restrictive when it might be becoming more liberal. This will occur when countries convert quotas into equivalent tariffs. But, as changes in the denominator (trade volume) reflect the net effect of all policy and exogenous changes, the index implicitly incorporates the effects of changes in all trade barriers.

8. A part of the increase in tax revenues could, partly, be attributed to fiscal "quick-fixes," i.e., proceeds from privatization are considered to be revenues.

9. A potential criticism of using this particular indicator to proxy trade openness is that it is not exogenous relative to the dependent variable. The problem could be remedied if instrumental variables, unrelated to trade taxes, are used to proxy the degree of trade openness. Lee (1993) and Frankel and Romer (1999) have attempted to develop indices of trade openness which have used instrumental variables such as geographical size, distance to major trading partners, the black market premium, etc. Rodriguez and Rodrik (1999) have however very convincingly demonstrated that these indicators of openness, that use instrumental variables, suffer from serious shortcomings and they conclude that "…simple averages of taxes on imports and exports… in fact do a decent job of rank-ordering countries according to the restrictiveness of their trade regimes" (p. 60).

10. When working with time-series data, one of the problems that could arise is the "spuriousness" of regression estimates if the time-series data is nonstationary (see, e.g., Macnair, Murdoch, Pi, & Sandler, 1995). But, given that the panel data set in this study has a relatively large number of cross-sections, interesting long-run relationships still exist between these integrated series. According to Phillips and Moon (1999), "these interesting relations are long-run average cross-section relationships… This makes sense in that the cross-section is usually assumed to reflect the long-run equilibrium relationship." The rationale that Phillips and Moon provide is that even if the noise in the time series within a cross-section is strong, it can be reduced by pooling the cross-sections. Thus, the result would be a consistent estimate of a long-run coefficient.

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APPENDIX A. COUNTRY CLASSIFICATION BY INCOME LEVEL

Low income	Lower middle income	Upper middle income	High income
Bangladesh	Colombia	Argentina	Australia
Bhutan	Costa Rica	Barbados	Austria
Burkina Faso	Dominican Republic	Botswana	Belgium
Burundi	Ecuador	Brazil	Canada
Cameroon	Egypt	Chile	Cyprus
Democratic Republic of Congo	Fiji	Malaysia	Denmark
Republic of Congo	Guyana	Malta	Finland
The Gambia	Jordan	Mauritius	France
Ghana	Morocco	Mexico	Greece
India	Papua New Guinea	Panama	Iceland
Kenya	Paraguay	Republic of Korea	Ireland
Lesotho	Peru	South Africa	Israel
Liberia	The Philippines	Uruguay	Italy
Madagascar	Sri Lanka		Japan
Malawi	Swaziland		Luxembourg
Myanmar	Syrian Arab Republic		The Netherlands
Nepal	Thailand		New Zealand
Nicaragua	Tunisia		Norway
Pakistan	Turkey		Portugal
Sierra Leone			Singapore
Uganda			Spain
Zambia			Sweden
Zimbabwe			Switzerland
			United Kingdom
			United Sates
23	19	13	25

Source: World Development Report 2000/2001.